

Nicholas K. Ung

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RELEVANT EXPERIENCE

Software Engineer, JT4

Sep 2025 – May 2026

- Enhanced operational ease by architecting a real-time Java Spring Boot transmission service integrating a modern UI.
- Achieved 99% code coverage and improved system reliability by migrating a key project's testing architecture to JUnit 5.
- Reduced manual execution time and enabled seamless enterprise-wide cross-service integration by designing and implementing a scalable RESTful API to replace a legacy CLI playback service.

Software Engineer Intern, JT4

May 2025 – Aug 2025

- Built a full-stack Vue, Java Spring Boot, and RabbitMQ application to interface with 8+ telemetry microservices.
- Prioritized low-latency data rendering by constructing a custom React time-series map plugin for a Grafana dashboard.
- Integrated an existing company project into an Azure DevOps CI/CD pipeline while writing over 60 robust tests in JUnit and Vitest to institute a test-driven development (TDD) approach.

Software Engineer (Part-Time), PolicyEngine

Mar 2023 – Mar 2025

- Drove a 15% increase in user engagement by engineering a responsive product dashboard with modern design patterns.
- Impacted the tax and benefit analyses of 100,000+ customers by delivering a novel API endpoint in a Flask backend.
- Overhauled the web app frontend using React, CSS, and JavaScript in close collaboration with the UI/UX team.

Software Engineer Intern, Amazon

Sep 2023 – Jan 2024

- Shipped a core product feature to a distributed system of 10,000+ users by taking ownership of the frontend React components for the OpenSearch Dashboards search relevance plugin.
- Protected long-term system health by increasing Jest test coverage by 11% and using CI/CD pipelines for rapid iteration.
- Ensured system stability under strict daily deadlines by performing rigorous root cause analysis and troubleshooting to resolve 12+ critical production bugs in TypeScript microservices.

Software Engineer (Contract), re:Bloom

May 2023 – Aug 2023

- Shipped a functional MVP in just two months by strategically bypassing custom builds and leveraging established integrations like PayPal.
 - Accelerated site traffic and search rankings by 20% by executing targeted SEO and performance optimization strategies.
 - Translated complex business requirements into delivered technical solutions by collaborating within a cross-functional Agile/Scrum environment.
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PROJECTS

PerfectRep | Python, Streamlit, OpenCV, rtmllib

- Developed an AI computer vision web application to analyze exercise form, utilizing pose estimation models to calculate real-time joint angles and depth metrics while providing actionable, customized feedback for users.
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TECHNICAL SKILLS

Languages: TypeScript, Java, Python, R, Go, C++, C, SQL, JavaScript, Bash, HTML/CSS

Frameworks & Libraries: Spring Boot, Vue, React, Angular, Next.js, Node.js, Express, Tailwind, Pytorch, Tensorflow

Testing & CI/CD: Vitest, JUnit 5, Jest, Azure DevOps, GitHub Actions

Infrastructure & Tools: Linux, Docker, RabbitMQ, AWS, GCP, Firebase, PostgreSQL, MySQL, MongoDB, Gradle, Git

EDUCATION

Master of Science - MS, Computer Science, Northeastern University

May 2026

- Coursework: Data Structures and Algorithms, Object Oriented Programming, Cloud Computing, Generative AI

Bachelor of Science - BS, Molecular, Cell and Developmental Biology, UCLA

Jun 2021

- Publications: Desai, Heta et al. "Multi-omic stratification of the missense variant cysteinome." bioRxiv : the preprint server for biology 2023.08.12.553095. 14 Aug. 2023, <https://doi.org/10.1101/2023.08.12.553095>.